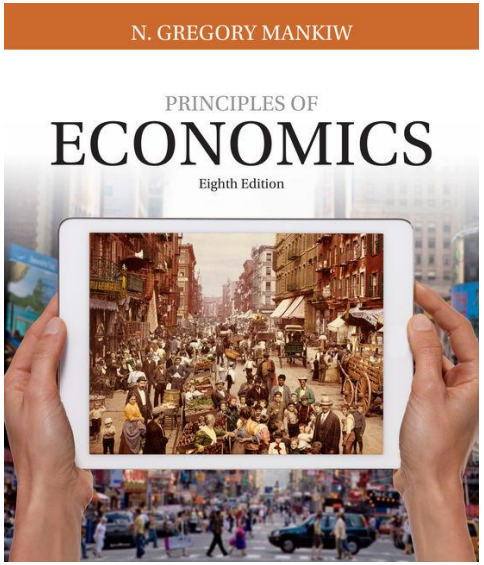
# IT3120- Modern Enterprise Management
# Introduction to Economics

Demand & Supply, Cost of Production, and Macroeconomics



# Learning Outcomes

- Introduction to Economics
- Supply and Demand
- The Costs of Production
- IT/SaaS Example: Cost of Production

# Concepts in Economics

# Needs and Wants

- **Needs**

Essentials required for survival

examples : food, shelter, clothing, and healthcare.

- **Wants**

Desires that improve quality of life but aren't essential

examples : smartphones, vacations, luxury goods.

# What is Economics?

"Economics is the study of how society manages its scarce resources."

## Society



Businesses



Governments



Individuals

**Manages**

Making decisions about production, consumption, and distribution

## Scarce Resources



Labor



Technology



Capital



Time



Land

# What is Economics?

Economics studies **how people and organizations make choices when resources are limited but needs and wants are unlimited**.

Since resources are limited, societies must decide:

- ✓ What goods and services to produce
- ✓ How to produce them
- ✓ Who should receive them



# Scarcity & the Economic Problem

## Finite Resources

Land, labor, and capital are limited in supply — they cannot be produced on demand or expanded infinitely.

## Infinite Wants

Human desires have no natural ceiling. The gap between what we have and what we want is the core economic problem.

## Opportunity Cost

Every choice carries a cost — the value of the next-best alternative foregone when a decision is made.



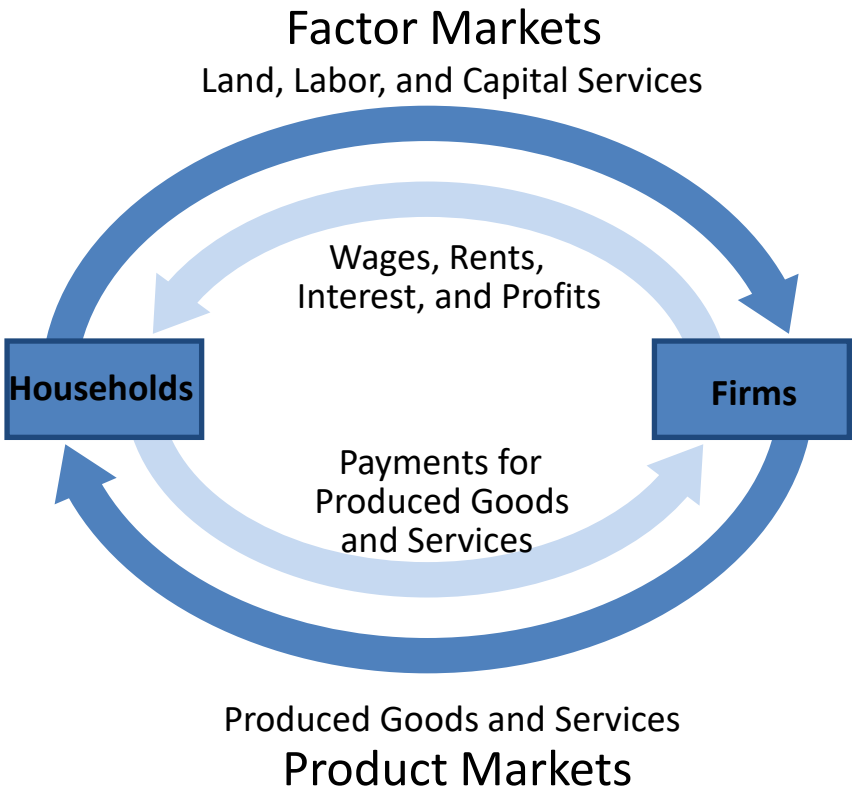
# Markets

- **Definition:**
  - **Market is a location (a place) where people go to buy and sell things**
  - **Physical or Virtual (i.e. eBay)**
  - **Bring buyers and sellers together.**

# Neoclassical Market

- A neoclassical market is a theoretical economic framework where prices, production, and consumption are driven by supply and demand;
    - **Supply & Demand:** Value of good/service is derived from perceived usefulness to the consumer rather than just the cost.
    - **Rational Behavior:** Buyers and sellers act logically and consistently, making choices that optimize their outcomes.
    - **Perfect Competition:** Assumes numerous buyers and sellers, identical products, and an absence of barriers to entry or exit, preventing any single entity from dictating prices.
    - **Market Equilibrium:** If there is a surplus or shortage, the market automatically adjusts prices until supply matches demand, resulting in the most efficient allocation of scarce resources.

# The Circular Flow Diagram for the Basic Neoclassical Model




- ✓Note: The neoclassical circular flow diagram represents a model in which there are only two kinds of economic actors, interacting through markets.

# Supply and Demand

# The Law of Demand

## The Inverse Relationship

As price rises, quantity demanded falls — and vice versa. Consumers respond rationally to changing prices.

## What Shifts Demand?

- Changes in consumer income
- Shifts in tastes and preferences
- Prices of related goods (substitutes & complements)
- Future price expectations



# Demand Curve




**Downward-sloping demand curve:** as price increases from low to high on the vertical axis, quantity demanded decreases from high to low on the horizontal axis.

## Each Point

Represents the **maximum price** consumers are willing to pay for a given quantity — their reservation price.



# Shifts in the Demand Curve

## Movement Along the Curve

Caused by a **change in price only**. The curve itself stays fixed — you simply move to a different point on the same line.

## Rightward Shift ↑

Demand **increases** at every price level.

Causes: rising incomes, positive advertising, population growth.

## Leftward Shift ↓

Demand **decreases** at every price level.

Causes: recession, cheaper substitutes, changing consumer tastes.

# Coffee vs. Tea: A Real Shift

When Starbucks raised prices by **10% in 2022**, demand for tea — a substitute good — rose noticeably across several markets. Consumers responded rationally to the relative price change.

The tea demand curve shifted **rightward** — not because tea's own price changed, but because a **related good's price did**. This is a non-price determinant shifting the entire curve.

# The Law of Supply

## Higher Prices → More Supply

As price rises, producers supply more — higher prices make production increasingly profitable, attracting more output.

## What Shifts Supply?

Input costs, technological improvements, the number of sellers in the market, and government policy all shift the supply curve.



# Supply Curve



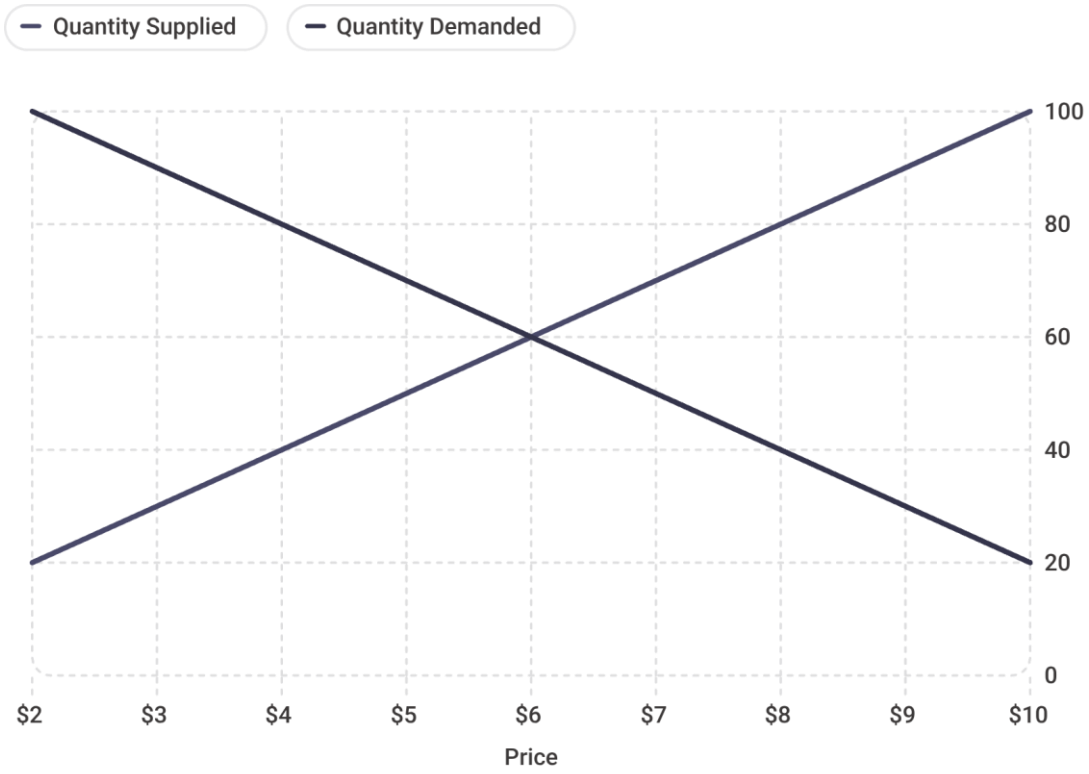

**Each Point**

Represents the minimum price producers require to supply a given quantity — effectively their marginal cost of production.

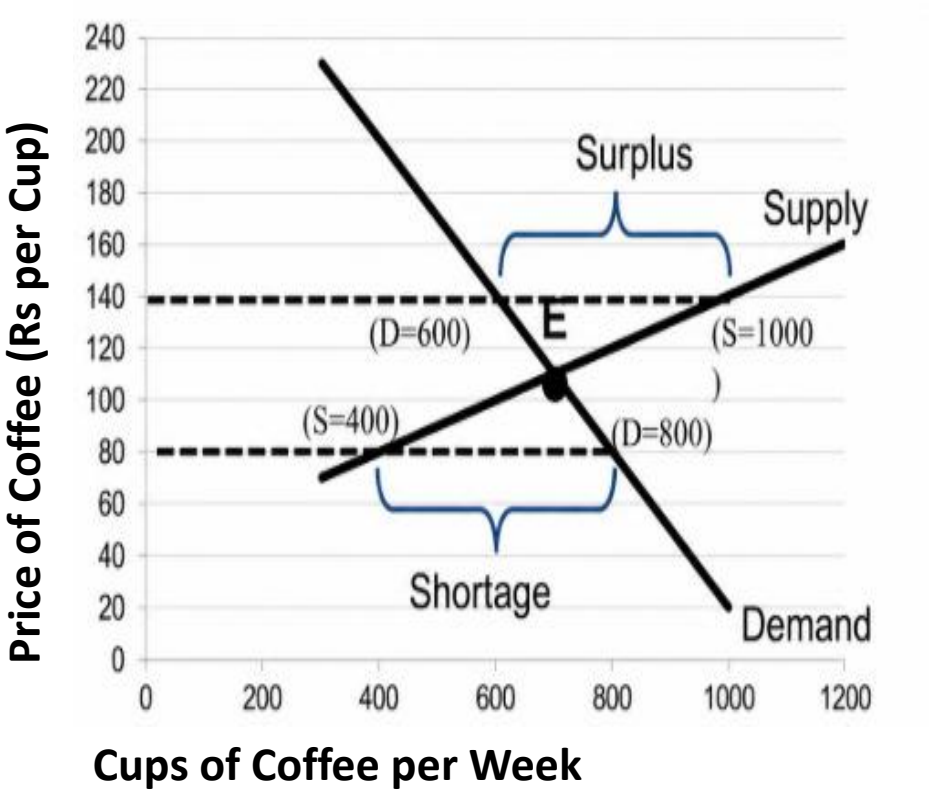


# Market Equilibrium

Equilibrium is the price at which quantity supplied equals quantity demanded — the market "clears." A surplus pushes prices down; a shortage pushes prices up.



# Surplus and Shortage

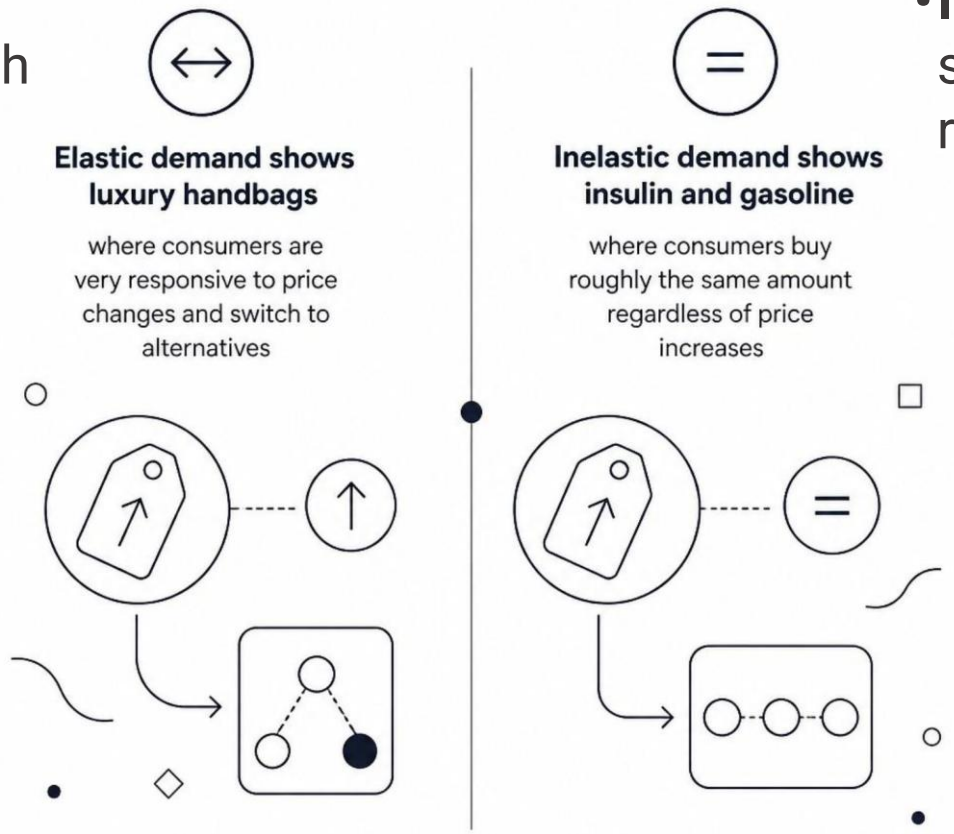



- At a price of Rs 140 a surplus occurs because the quantity of cups of coffee being offered for sale is larger than the quantity that people want to buy.
- On the other hand, at a price of Rs 80 many people want to buy cups of coffee but few want to sell, so a shortage occurs.
- Only at market equilibrium (point E) does quantity supplied equal quantity demanded

# Elasticity: How Sensitive Are Buyers?

Elasticity measures how strongly buyers respond to price changes. It determines how firms price products and how taxes ripple through markets.

- **Elastic:** demand drops sharply with price hikes




- **Inelastic:** demand stays stable regardless of price

# Cost of Production

# Costs

## Fixed Costs:

- Remain constant regardless of whether a firm produces a large amount of output or produces nothing at all.
- Associated with resources that cannot be easily adjusted in the short run.

## Variable Costs:

- As a firm increases its production, variable costs increase, and when production decreases, variable costs also decrease.
- Directly related to the resources used in the production process, such as labor, raw materials, and other inputs that vary with the level of production.

## Marginal cost:

- Refers to the increase in total cost that arises from producing one additional unit.
- Helps firms determine how production changes affect their overall costs.
- By comparing the marginal cost of additional unit with the potential benefit, firms can decide the most efficient level of production.

# Cost of Production

- Production costs are all the direct and indirect expenses a business incurs to manufacture a good or provide a service. They are divided into fixed costs (rent, insurance) and variable costs (raw materials, labor).

$$\text{Cost Per Unit} = \frac{\text{Total Fixed Costs} + \text{Total Variable Costs}}{\text{Total Units Produced}}$$

  - Variable Costs (scale with production volume)
    - Raw materials
    - Direct labor
  - Fixed Costs (remain steady regardless of output)
    - Factory rent
    - Equipment maintenance/depreciation

# Opportunity Cost



**Opportunity cost** refers to **the value of the next best alternative that must be given up when a choice is made**.



Opportunity cost highlights the idea that **every decision involves giving up an alternative**, and rational decision-making requires considering the value of the forgone option.

# Opportunity Cost Example

- Consider a **software development company** that has a limited number of developers and must decide how to allocate them. The company may choose between:
  - Developing a **new mobile application**, or
  - **Improving an existing software product**

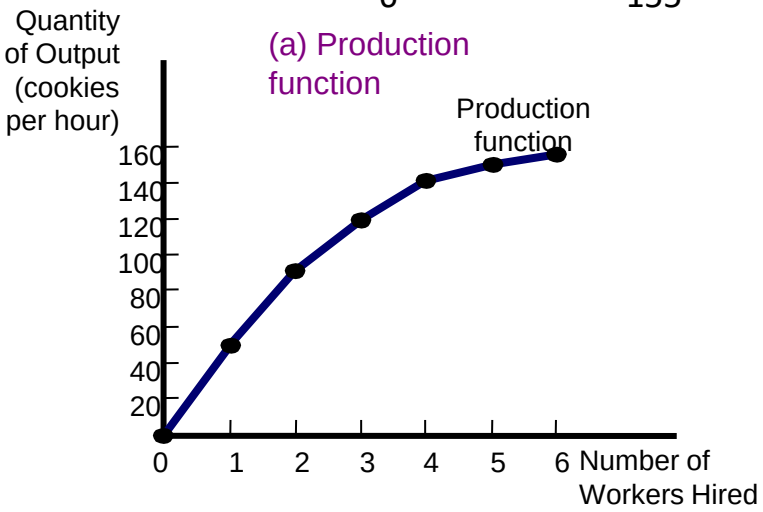
- If the company decides to allocate its developers to create the new mobile application, it cannot simultaneously use those developers to enhance the existing product. In this case, the **opportunity cost** of developing the new application is the potential benefits that could have been gained from improving the existing software.

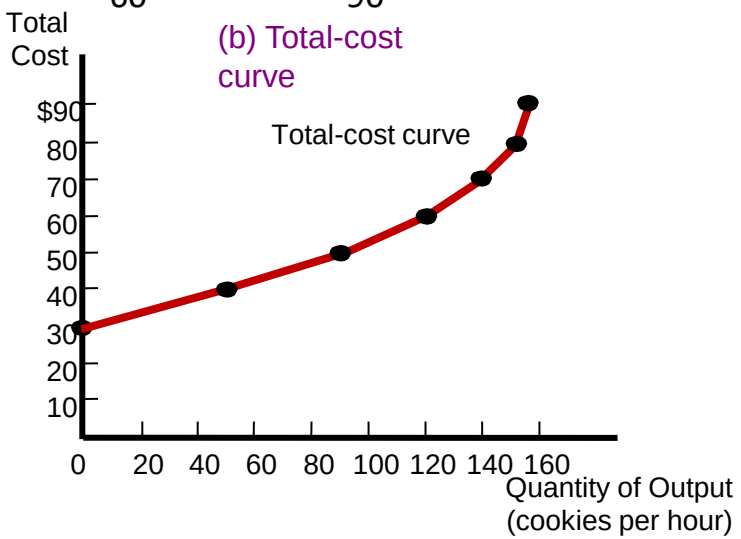
# Measures of Cost

- Fixed costs
  - Do not vary with the quantity of output produced
- Variable costs
  - Vary with the quantity of output produced
- Average fixed cost (AFC)
  - Fixed cost divided by the quantity of output
- Average variable cost (AVC)
  - Variable cost divided by the quantity of output

# E.g. Production function and Total-cost curve

| Number of workers | Output (Cookies/ per hr) | Marginal product of labor | Cost of factory | Cost of workers | Cost of inputs (cost of factory + cost of workers) |
|---|---|---|---|---|---|
| 0 | 0 | | $30 | $0 | $30 |
| 1 | 50 | 50 | 30 | 10 | 40 |
| 2 | 90 | 40 | 30 | 20 | 50 |
| 3 | 120 | 30 | 30 | 30 | 60 |
| 4 | 140 | 20 | 30 | 40 | 70 |
| 5 | 150 | 10 | 30 | 50 | 80 |
| 6 | 155 | 5 | 30 | 60 | 90 |



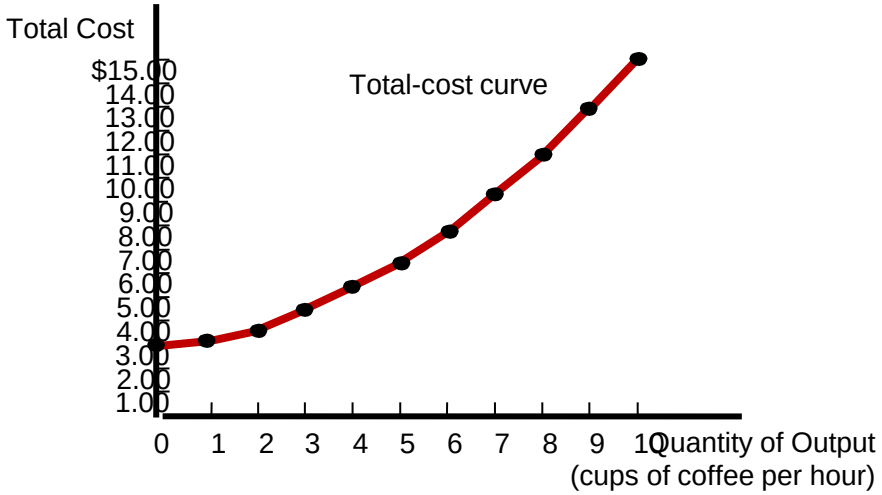

The production function gets flatter as the number of workers increases, which reflects diminishing marginal product.

The total-cost curve gets steeper as the quantity of output increases because of diminishing marginal product.

# Measures of cost: Coffee shop

| Quantity of coffee (cups/hr) | Total Cost | Fixed Cost | Variable Cost | Avg Fixed Cost | Avg Variable Cost | Avg Total Cost | Marginal Cost |
|---|---|---|---|---|---|---|---|
| 0 | $3.00 | $3.00 | $0.00 | - | - | - | |
| 1 | 3.30 | 3.00 | 0.30 | $3.00 | $0.30 | $3.30 | $0.30 |
| 2 | 3.80 | 3.00 | 0.80 | 1.50 | 0.40 | 1.90 | 0.50 |
| 3 | 4.50 | 3.00 | 1.50 | 1.00 | 0.50 | 1.50 | 0.70 |
| 4 | 5.40 | 3.00 | 2.40 | 0.75 | 0.60 | 1.35 | 0.90 |
| 5 | 6.50 | 3.00 | 3.50 | 0.60 | 0.70 | 1.30 | 1.10 |
| 6 | 7.80 | 3.00 | 4.80 | 0.50 | 0.80 | 1.30 | 1.30 |
| 7 | 9.30 | 3.00 | 6.30 | 0.43 | 0.90 | 1.33 | 1.50 |
| 8 | 11.00 | 3.00 | 8.00 | 0.38 | 1.00 | 1.38 | 1.70 |
| 9 | 12.90 | 3.00 | 9.90 | 0.33 | 1.10 | 1.43 | 1.90 |
| 10 | 15.00 | 3.00 | 12.00 | 0.30 | 1.20 | 1.50 | 2.10 |

The total-cost curve gets steeper as the quantity of output increases because of diminishing marginal product.



# Measures of Cost

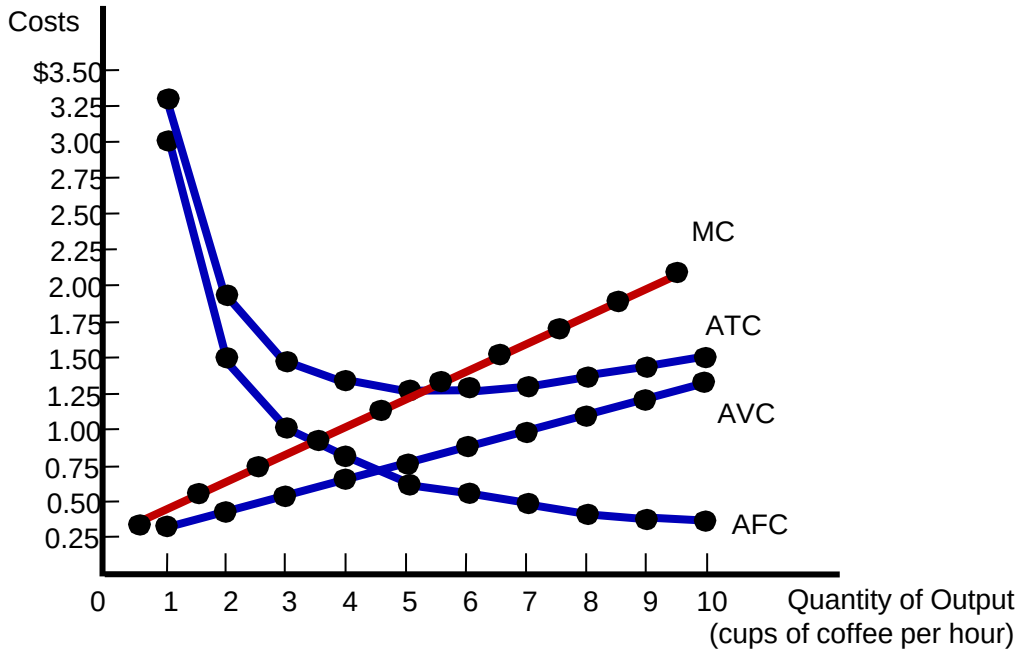
- Average total cost (ATC)
  - Total cost divided by the quantity of output
  - Average total cost = Total cost / Quantity

    $ATC = TC / Q$

- Marginal cost (MC)
  - Increase in total cost
    - Arising from an extra unit of production
  - Marginal cost = Change in total cost / Change in quantity

    $MC = \Delta TC / \Delta Q$

# Average-cost & Marginal-cost Curves



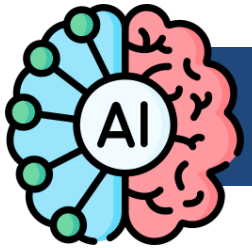

This figure shows the average total cost (ATC), average fixed cost (AFC), average variable cost (AVC), and marginal cost (MC) for Conrad's Coffee Shop. (Data in Table 2).

These cost curves show three features that are typical of many firms:
(1) Marginal cost rises with the quantity of output.
(2) The average-total-cost curve is U-shaped.
(3) The marginal-cost curve crosses the average-total-cost curve at the minimum of average total cost.

# Measures of Cost

- Cost curves and their shapes
- U-shaped average total cost: ATC = AVC + AFC
  - AFC – always declines as output rises
  - AVC – typically rises as output increases
    - Diminishing marginal product
  - The bottom of the U-shape
    - At quantity that minimizes average total cost

- Relationship between MC and ATC
  - When MC < ATC: average total cost is falling
  - When MC > ATC: average total cost is rising
  - The marginal-cost curve crosses the average-total-cost curve at its minimum




# Using AI for Cost Calculations

- AI can enhance how firms calculate and manage different types of production costs by **automating data analysis, improving accuracy, and supporting decision-making**.

- **Cost of Production -** AI can analyze large datasets (labor, materials, infrastructure) to estimate the **total cost of production**. It can also identify inefficiencies and suggest ways to reduce overall costs.

➢ **Fixed Cost -** AI helps track fixed costs such as salaries, rent, and infrastructure. It can allocate fixed costs across different products or projects to better understand cost distribution.

- **Variable Cost -** AI can monitor how costs change with output levels (e.g., cloud usage, raw materials). It predicts how variable costs will increase or decrease as productionscales.

➢ **Marginal Cost -** AI can estimate the **additional cost of producing one more unit** by analyzing incremental changes in resources and output. This helps firms decide the optimal level of production.

# Benefits



**Benefits** refer to the **gains or satisfaction (utility)** that consumers or producers receive from consuming or producing goods and services.



Benefits help in understanding how efficiently resources are used and how well a market satisfies the needs of society.

# Benefits Calculation

- **Consumer Benefits**
  - Consumers benefit when they are willing to pay more for a product than the actual price.
  - This extra value is known as **consumer surplus**.

**Consumer Benefit = Willingness to Pay – Actual Price**

- **Producer Benefits**
  - Producers benefit when they sell a product at a price higher than their cost of production.
  - This gain is known as **producer surplus**.

**Producer Benefit = Price Received – Cost of Production**

- **Total Economic Benefit**
  - Total benefit in a market is the **sum of consumer surplus and producer surplus**.
  - It reflects the **overall welfare or efficiency** of the market.

# Benefits Calculation Example

- **Consumer Side:**

  A student is willing to pay $50 for software but buys it for $30

  **Consumer Benefit = Willingness to Pay – Actual Price**

  **= $50 - $30 = $20**

- **Producer Side:**

  A firm produces a product at a cost of $20 and sells it for $30

  **Producer Benefit = Price Received – Cost of Production**

  **= $30 - $20 = $10**



# In Class Activity 1

# Multiple Choice Question

**Opportunity cost refers to:**

A. Total cost of production

B. The next best alternative forgone

C. Monetary cost only

D. Fixed cost



# Short Answer Question

A software company can either develop a **new mobile app** or **improve an existing system**.

What is the **opportunity cost** of choosing to develop the new app?





# In Class Activity 2

# Multiple Choice Question

**According to the law of demand:**

A. Price ↑ → Demand ↑

B. Price ↓ → Demand ↓

C. Price ↓ → Demand ↑

D. Demand is constant



# Multiple Choice Question

**Which of the following increases supply?**

A. Increase in production cost

B. Improvement in technology

C. Decrease in number of sellers

D. Higher taxes



# Short Answer Question

During a big online sale, demand for a software service increases suddenly.

What is likely to happen to the **price** if supply remains constant?





# In Class Activity 3

# Multiple Choice Question

**Fixed cost:**

A. Changes with output

B. Remains constant regardless of output

C. Depends on demand

D. Remains always zero

